Act as an expert academic researcher and technical writer in the fields of Artificial Intelligence, Computer Vision, and Industrial Engineering. Your task is to write a comprehensive, highly technical first draft of a research paper intended for publication in a Scopus-indexed conference.

Project Context:

Our research solves a major bottleneck in process industries: the manual extraction of Electrical and Instrumentation (E&I) deliverables (Instrument Index, I/O List, Alarm List, Trip List) from Piping and Instrumentation Diagrams (P&IDs). Current AI approaches suffer from a "Contextual Integrity Gap"—they use object detection to find symbols but fail to understand functional engineering logic. To solve this, our system uses a multi-layered Hybrid AI architecture:

Perception Layer: YOLOv8 for symbol bounding-box detection, and OCR (EAST/Tesseract) for alphanumeric tag extraction.

Topological Layer: NetworkX and Neo4j to map physical connectivity into a graph.

Deterministic Layer: Hard-coded Python algorithms that strictly validate extracted data against ISA-5.1 (instrument tagging) and IEC 61511 (functional safety trip logic) standards.

Cognitive Layer: Graph-based Retrieval-Augmented Generation (Graph-RAG) using LangChain and LLMs to semantically reason over the validated graph, completely eliminating AI hallucinations.

Output Layer: Pandas to generate the final structured Excel spreadsheets.

Draft Requirements:

Please generate the paper using the exact structure below. Maintain an objective, formal, and highly academic tone throughout.

1. Title: Extraction of E&I Deliverables from P&ID using Generative AI.

2. Abstract: Write a strict 200-word abstract. Summarize the industrial problem, the failure of purely probabilistic computer vision models, our proposed hybrid deterministic-RAG architecture, and the impact of automating E&I document generation for the EPC sector.

3. Introduction: Detail the industrial context. Explain what P&IDs are, why manual data extraction is dangerous and costly, and define the "Contextual Integrity Gap." State the primary objective of this research.

4. Literature Review: Map the evolution of P&ID digitization across 4 stages:

Stage 1: Classical rule-based image processing.

Stage 2: Deep learning object detection (CNNs, YOLO).

Stage 3: Graph-based topological modeling and OCR limitations.

Stage 4: Emerging Vision-Language Models.

Conclude this section by clearly identifying the research gap: the lack of embedded ISA-5.1 and IEC 61511 engineering rules combined with generative AI.

5. Methodology: Provide a deep technical breakdown of our proposed system architecture. Detail the data flow from the raw PDF input, through the YOLO/OCR perception layer, into the Neo4j graph database. Explain how the deterministic rule engine validates tags, and specifically detail how Graph-RAG is utilized to safely apply LLM reasoning to engineering data without hallucinating.

6. Results & Discussion: Write this section assuming a highly successful proof-of-concept prototype. Discuss how the integration of Graph-RAG and deterministic rules eliminated the false positives normally seen in pure vision models. Mention high bounding-box recall rates and zero-violation compliance with ISA-5.1 naming conventions. (Please leave clearly marked placeholders like or for data we will add later).

7. Conclusion: Provide a strong summary of how this hybrid architecture successfully bridges the gap between raw pixel detection and semantic engineering intent, paving the way for safer and faster industrial automation. Discuss future scope (e.g., training custom YOLO models on larger datasets).